MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) — FOURTH SEMESTER - March 2026

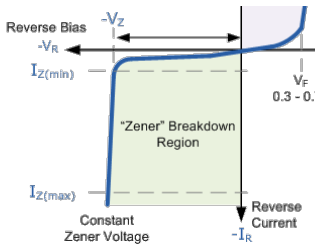
MG4DSCPHY202 — BASIC ELECTRONICS AND ELECTRICITY

SCHEME OF VALUATION

Maximum Marks: 50 - Duration: 1 hour 30 minutes

PART A — Short Answer Questions (Each carries 2 marks - Attempt any 10)

Q. No	Question	Answer / Key Points	Marks	CO	LD
1	Define 1 Volt	One Volt is defined as the potential difference between two points in a conductor such that 1 Joule of work is done in moving 1 Coulomb of charge from one point to the other. Formula: V = W/Q (1 V = 1 J/C)	2 marks for definition; (1 mark for formula alone)	CO1	K
2	Write the expression for instantaneous EMF generated in an AC generator. What will be the EMF when the coil is perpendicular to the field?	Instantaneous EMF: $e = E_0 \sin(\omega t)$, where E_0 = peak EMF, ω = angular frequency. When the coil is perpendicular to the field, ωt= 0°. $\therefore e = E_0 \sin(0°) = 0$. - Alternate expressions in *cos* or *exp* shall also be given credits.	1 mark for expression; 1 mark for EMF value	CO1	U
3	State Lenz's Law	Lenz's Law: The direction of the induced EMF (and hence induced current) is always such that it opposes the cause that produces it. Or $e = -N\frac{d\Phi}{dt}$ It is a consequence of the law of conservation of energy.	2 marks for statement (1 mark for significance)	CO1	U
4	What is the effect of temperature on electrical resistance of insulators?	For insulators, resistance decreases with increasing temperature.	1 mark	CO1	U
		At higher temperatures, more charge carriers are freed (valence electrons gain sufficient energy to jump the band gap), so conductivity increases and resistance decreases. OR Insulators have a negative temperature coefficient of resistance.	1 mark		
5	What is the relation between voltage across resistor and voltage across inductor in an LR circuit?	In an LR series circuit, $V_R = iR$ (in phase with current) and $V_L = iX_L$ or $V_L = L\frac{di}{dt}$ (leads current by 90°). So V_R and V_L are 90° out of phase with each other. OR figure depicting the same. Total voltage $V = \sqrt{(V_R^2 + V_L^2)}$	2 marks	CO2	U

Q. No	Question	Answer / Key Points	Marks	CO	LD
6	Obtain ripple factor for a half-wave rectifier.	For half-wave rectifier: $I_{rms} = I_m/2$, $I_{dc} = I_m/\pi$	1 mark	CO3	U
		Ripple factor $\gamma = \sqrt{\left[(I_{rms}/I_{dc})^2 - 1\right]} = \sqrt{\left[(\pi/2)^2 - 1\right]} = 1.21$	1 mark		
7	Sketch and explain the reverse characteristic of a Zener diode.	In reverse bias, a small reverse saturation current flows until the reverse voltage reaches the Zener breakdown voltage V_Z. At V_Z, the current increases sharply (avalanche/Zener breakdown) while voltage remains nearly constant at V_Z. This constant voltage property is utilized for voltage regulation. 	1 mark for reverse characteristic 1 mark for explanation	CO3	U
8	What are the properties of a PN junction?	1. The Depletion Layer and Barrier Potential 2. Biasing Characteristics 3. Unidirectional Nature 4. V-I Characteristic Curve	2 marks for any 2 properties	CO3	U
9	Give reasons for the permanent destruction of the PN juncion with high reverse voltage.	The permanent destruction of a PN junction when subjected to a high reverse voltage occurs due to a phenomenon known as reverse breakdown. A combination of a sharp increase in current and the resulting intense heat generation leads to the physical and electrical failure of the semiconductor device.	2 marks	CO3	U
10	Why is the current gain of a CE configuration greater than unity?	In CE configuration, $\beta\,(current\ gain) = I_C/I_B$. Since $I_C \gg I_B$ (as emitter is heavily doped and collector collects most injected carriers), β >> 1. Typical β values range from 20 to 500.	1 mark for β definition; 1 mark for correct reasoning	CO4	U
11	What is meant by knee voltage in a common-emitter transistor configuration?	Knee voltage (or saturation voltage) is the minimum collector-emitter voltage below which the collector current falls sharply even for a given base current. Below knee voltage, the transistor operates in saturation region.	2 marks	CO4	U
12	Explain how transistor amplifies a weak signal?	A weak signal applied to the base-emitter (low resistance) junction causes small variations in base current ΔI_B. Due to transistor action, $\Delta I_C = \beta \times \Delta I_B$, producing large collector current variation. This large current change through the high-resistance collector load produces a large output voltage, achieving amplification.	2 marks (Expression alone 1 mark)	CO4	A
13	List the characteristics of an ideal Operational Amplifier.	1. Infinite open-loop gain, 2. Infinite input impedance, 3. Zero output impedance 4. Infinite bandwidth, 5. Zero offset voltage ($V_{out} = 0$ when $V_{input} = 0$) 6. Infinite Common-Mode Rejection Ratio (CMRR = ∞), 7. Infinite Slew Rate	Any 4 characteristics - 2 marks	CO5	U
14	What is an inverting operational amplifier?	In an inverting op-amp, the input signal is applied to the inverting (-) terminal through resistor R_{in}, and the non-inverting (+) terminal is grounded.The output is 180° out of phase with the input. Voltage gain $A_v = -R_f / R_{in}$	1 mark for input terminal description; 1 mark for gain formula	CO5	U

PART B — Short Essay Questions (Each carries 5 marks - Answer any 6)

Q.No	Question	Answer / Key Points	Marks	CO	LD
15	The resistance of a conductor is 12 Ω at 0°C and 14.4 Ω at 50°C. Find the temperature coefficient of resistance of the material.	Formula: $R_t = R_0(1 + \alpha t)$ $14.4 = 12(1 + \alpha \times 50)$ $14.4/12 = 1 + 50\alpha$ $1.2 = 1 + 50\alpha$ $\alpha = 0.2/50 = 0.004$ /°C Temperature coefficient of resistance = 4×10^{-3} /°C	2 marks for formula, 2 marks for substitution and final answer 1 marks for unit.	CO1	U
16	An ac current of frequency 50Hz is applied through a coil containing 200 mH. Calculate the inductive reactance offered to the flow of ac current.	Formula: $X_L = 2\pi fL$ $X_L = 2 \times 3.14159 \times 50 \times 200 \times 10^{-3}$ $= 2 \times 3.14159 \times 50 \times 0.2$ $= 62.83\ \Omega$	2 marks for formula, 2 marks for substitution and final answer 1 marks for unit.	CO2	U
17	Derive the expression for current in an LCR series circuit and discuss its variation with frequency.	Applied voltage: $V = V_m \sin(\omega t)$ Impedance: $Z = \sqrt{[R^2 + (X_L - X_C)^2]}$ Current: $I = V/Z = V_m/Z \sin(\omega t - \varphi)$ Phase angle: $\tan\varphi = (X_L - X_C)/R$ Frequency variation: At low f: $X_C > X_L$, circuit is capacitive, I lags V. At resonance $f_0 = 1/(2\pi\sqrt{LC})$: $X_L = X_C$, Z = R (minimum), I is maximum. At high f: $X_L > X_C$, circuit is inductive, I leads V.	2 marks for derivation; 1 mark for phase angle; 2 marks for discussion on frequency variation.	CO2	U
18	Explain the mechanism by which a Zener diode keeps the output voltage constant when the input supply voltage varies.	The Zener diode is connected in reverse bias with a series resistor R_s. When input voltage V_{in} increases: extra current flows through R_s; excess current is diverted through Zener, so V_{out} remains constant at V_Z. When V_{in} decreases: Zener current decreases, less voltage drops across R_s, V_{out} stays at V_Z. The Zener operates in the breakdown region where V−I curve is nearly vertical, maintaining constant voltage.	5 marks (Circuit alone 2 marks)	CO3	U
19	A transformer supplies 30 Vrms to a full-wave rectifier. Find the maximum voltage and dc output voltage of a	Given: $V_{rms} = 30$ V Maximum voltage: $V_m = V_{rms} \times \sqrt{2} = 30 \times 1.414 = 42.43$ V For bridge rectifier, DC output voltage: $V_{dc} = 2V_m/\pi = 2 \times 42.43 / 3.14 = 27.02$ V	2 marks for formula, 2 marks for substitution and final answer 1 marks for unit.	CO3	A

Q.No	Question	Answer / Key Points	Marks	CO	LD
	bridge rectifier.				
20	Compare the output voltage variation in a rectifier circuit without a filter and with a capacitor filter.	Without filter: Output is pulsating DC with large ripple. Voltage rises to V_m then falls to zero (half-wave) or V_m then to zero (full-wave) at double frequency. With capacitor filter: The capacitor charges to peak voltage V_m during conduction and slowly discharges through load between peaks. Include comparative plots	Description + plots – 5 marks Plots or Description alone - 2.5 marks	CO3	An
21	Describe with circuit diagram, input and output characteristics of a CE amplifier.	Circuit: Common emitter configuration — emitter is common to both input (base-emitter) and output (collector-emitter). Input Characteristics (I_B vs V_{BE} at constant V_{CE}): • Similar to forward biased diode curve. • I_B increases exponentially with V_{BE} beyond ~0.6–0.7 V (Si). Output Characteristics (I_C vs V_{CE} at constant I_B): • Three regions: Active, Saturation, Cut-off. • Family of curves for different I_B values; spacing shows current gain β.	Description – 2 marks Circuit diagram – 1 marks Input characteristics – 1 mark output characteristics – 1 mark	CO4	U
22	Define an operational amplifier and explain its significance in electronic circuits.	An Operational Amplifier (Op-Amp) is a high-gain differential amplifier with very high input impedance, very low output impedance, and the ability to amplify AC and DC signals. Key parameters: Open-loop gain ~10^5 to 10^7, CMRR > 90 dB, Bandwidth up to MHz range. Significance: 1. Versatile: Used as amplifier, integrator, differentiator, comparator, oscillator, filter. 2. Closed-loop feedback allows precise gain control: $A_v = -R_f/R_{in}$. 3. High CMRR rejects noise effectively. 4. Compact (IC form — e.g., μA741) and inexpensive. 5. Foundation of analog signal processing.	Description - 3 marks Significance – 2 marks	CO5	U